



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Mid-Spring Semester Examination 2023-24

Date of Examination: 19.02.24 Session: AN Duration: 2 hrs Full Marks: 40

Subject No.: AI60004 Subject: Big Data processing

Dept/Center/School: Centre of Excellence in Artificial Intelligence

Specific charts, graph paper, log book required: NO

(Answers without detailed justification won't be given any credit.)

1. Answer the questions below. 5x2

- a. In a datacentre what role do the collocation suites play? What are they consist of?
- b. Rank (in descending order) the following memory devices based on their latency:
Cluster disk, rack disk, Rack L2 Cache, Cluster DRAM, rack DRAM.
- c. You are writing 1 GB of data into HDFS. The block size is 128 MB. What is the
 $1024/128 = 8$ maximum number of datanodes where the data might be written? Assume that the
 $8*3 = 24$ cluster has 10000 datanodes and the replication factor is 3.
- d. What is HDFS federation? What is its necessity?
- e. When do you prefer scale-up system over scale-out system? Give an example.

5+5

2. a) The map phase of a map-reduce program produces 1000 key-value pairs with 10 distinct keys and running on a cluster of 10 datanodes. You are using a partition function f such that $f(a)=f(b)$ if and only if $a=b$. Assume that the M-R system enforces the best load balancing mechanism for the data distribution.
- i) In the worst case, what is the maximum and minimum number of values a reduce function has to process? Explain.
 - ii) If a node running a reduce task fails during middle of the computation, how does the map-reduce framework address the issue? Write this in steps (Step 1, Step 2, ...).

b) What is data race in parallel programming? Give an example code (in pseudo code) to illustrate data race. Suggest a suitable mechanism to address non-determinism in your example code.

5+5

3. a) A company has a distributed cluster with n datacentres ($d_1, d_2, \dots, d_n$), where each datacentre has m racks and each rack has k datanodes. Each backbone switch of the cluster can connect at most f switches to its immediate next level (below).
- i) Compute the minimum number of backbone switches that the company has to use.
 - ii) Assuming that the number of backbone switches are minimum, compute the maximum possible network distance between two data nodes in the cluster.

b) You are given m d -dimensional data points (all real valued). You need to cluster the data points into k clusters using k -means clustering algorithm in map-reduce programming framework. Write the map function for this. You must specify input key-value, output key-value and the body of the map function. Use pseudocode to write the function. (Textual description won't be given any credit.)

4. A big matrix M is represented as a set of sub-matrices as follows:

$M = \begin{pmatrix} S_{11} & \cdots & S_{1n} \\ \vdots & \ddots & \vdots \\ S_{n1} & \cdots & S_{nn} \end{pmatrix}$, where each submatrix is of size 2×2 . We transform the matrix M to create a new matrix M' such that the (i,j) -th element of M' , $m'_{ij} = S_{ij} * K$, where K is a 2×2 matrix. The operation $A * B$ between two 2×2 matrices A and B is defined as: $a_{11} * b_{11} + a_{12} * b_{12} + a_{21} * b_{21} + a_{22} * b_{22}$.

- a) Write a multi-threaded algorithm for the above problem. The algorithm should be in pseudo code. Assume that the number of threads is m and n is multiple of m .
- b) We redefine the operation such that the (i,j) -th element of M' , $m_{ij} = c \times (S_{ij} * K)$, where $c = 1$ if all $S_{pq} * K$ are positive and $c = 0$ otherwise for $(p < i, q < j)$. Modify the above algorithm to implement the new operation. The algorithm must be multithreaded.